Detecting Nearshore Underwater Targets With Hyperspectral Nonlinear Unmixing Autoencoder

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***Abstract*— Hyperspectral underwater target detection (HUTD) is a promising and challenging task in remote sensing image processing. Existing methods face significant challenges when adapting to nearshore environments, where cluttered backgrounds hinder the extraction of target signatures and exacerbate signal distortion. Hyperspectral unmixing (HU) demonstrates potential effectiveness for nearshore underwater target detection (UTD) by simultaneously extracting water background endmembers and separating target signals. To this end, this article investigates a novel nonlinear unmixing network for hyperspectral UTD, denoted as nonlinear unmixing network for hyperspectral-UTD (NUN-UTD), in which a well-designed autoencoder-based unmixing network is used to obtain the abundance map as the detection result. To address the weak underwater target signals, a target prior spectral preservation scheme is employed to guide the unmixing network in learning the accurate target abundance. Besides, to address the complexity of the nearshore environment, a pseudomixed data classification constraint is incorporated into the objective function to enhance the discriminative capability between the background and the target. Moreover, we adopt an additive postnonlinear model in the decoder to deal with the interactions between underwater spectra to account for the nonlinear effects between spectra of underwater substances. To validate the effectiveness of the proposed method, we constructed a hyperspectral dataset for nearshore UTD. Extensive experiments conducted on three real-world datasets and one simulated dataset demonstrate that our method achieves outstanding performance in HUTD.**

***Index Terms*— Autoencoder, hyperspectral image (HSI), nearshore underwater target detection (UTD), unmixing network, weak target signals.**

I. Introduction

THE critical technologies of underwater target detection (UTD) hold significant potential and promising prospects for applications in marine resource exploration and ecological monitoring, attracting increasing attention from a growing number of scholars [1], [2], [3], [4], [5]. Hyperspectral images (HSIs) are renowned for their extensive and continuous spectral bands, offering a wealth of spectral information that enables the fine-grained identification of diverse materials [6], [7]. This characteristic provides considerable potential for the detection and recognition tasks in complex scenarios. Specifically, in remote sensing of underwater scenes, the acquisition of underwater target contour information from standard RGB images is significantly impeded due to the absorption and refraction of light by water bodies. This presents a considerable challenge for UTD. In comparison, hyperspectral imagery captures both the spatial texture of the scene and a wealth of spectral information, making it more suited for UTD. Therefore, hyperspectral UTD (HUTD), as a novel underwater reconnaissance technology, has been extensively studied across various domains, such as oceanography and marine biology in recent years [8], [9], [10].

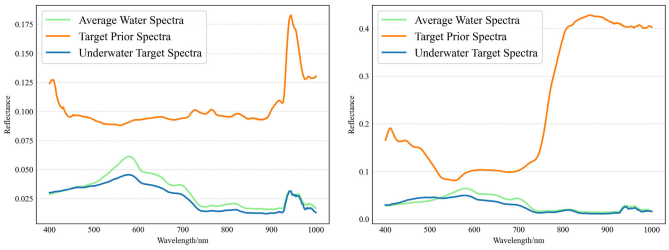


Fig. 1. On the River Scene datasets, a comparison is presented between the average water spectrum (green curve), the target prior spectrum (yellow curve), and the underwater target spectrum (blue curve). The target prior spectrum was collected in advance from the shore, while the underwater target spectrum was obtained by averaging the spectra at locations with ground truth. (a) River Scene 1. (b) River Scene 2.

Traditional hyperspectral target detection (HTD) algorithms typically rely on matching the target prior spectra with the spectral signatures of image pixels [11], [12], [13], [14], [15] or utilize anomaly detection methods to identify targets as outliers [16], [17], [18]. However, anomaly detection and conventional HTD algorithms might fail to discriminate targets in underwater imagery since the target prior spectra undergo significant distortion. When imaging underwater targets, the light traverses the water body twice, getting absorbed and scattered. As shown in our River Scene datasets (see Fig. 1), the underwater target spectra exhibit significant differences from the target prior spectrum. Consequently, the received underwater

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target spectrum highly resembles the water spectrum, making it difficult to discriminate. Traditional algorithms, thus, suffer severe performance degradation in underwater scenarios. Therefore, the primary objective of HUTD is to overcome the distortions in target spectra caused by water environments and detect underwater targets of interest.

HUTD employs preobtained target prior spectra to detect underwater targets within HSI. To address the distortion in target spectra, numerous HUTD approaches have been developed and can be categorized into two primary categories. One is the water inherent optical property (IOP) inversion-based approach. In this approach, water IOPs are first estimated [19], [20], [21] and then incorporated into a bathymetric model [22] to determine underwater target depth and recover the target spectra. The recovered target spectra are utilized as references for conventional methods to detect underwater targets. Jay et al. [23] adapted the traditional detection algorithms into bathymetric versions to account for spectral distortions induced by the water column. Furthermore, they proposed a novel statistical approach [24] that leverages neighboring pixel information to enhance robustness against noise. It improves the accuracy of mapping depth and water quality parameters across varying water depths. Qi et al. [25] proposed an iterative joint depth estimation and target detection network for UTD, which can refine itself by incorporating both spectral and spatial information in an iterative manner. Li et al. [26] leverage synthetic data generated from the bathymetric model to train deep learning models and subsequently apply them to realistic data. Li et al. [27] introduced 3-D convolutions and a depth estimation network to enhance the detection performance. Although these IOPs' inversion-based methods have demonstrated remarkable detection performance, estimating water IOPs and depth is challenging in nearshore environments with turbid water and depth variations. Moreover, the detection performances of these algorithms hinge critically on the precision of estimated IOPs and bathymetric modeling.

The other one is the signal separation-based approach utilizing hyperspectral unmixing (HU). Over the past decades, autoencoder-based HU has been extensively studied with deep learning techniques [28], [29], [30], [31], [32]. Meanwhile, there also have been numerous outstanding works on HU based on nonlinear autoencoders [33], [34], [35], [36], [37], [38]. Zhu et al. [39], [40] introduced the concept of abundance learning, emphasizing the collaborative term among endmembers to achieve accurate abundance learning, facilitating the purpose of target detection. Qi et al. [41] pioneered HU for UTD, presuming only water and target per scene. Compared to the IOPs' inversion-based approaches, HU might be better suited for nearshore UTD because HU can extract target signals and separate the endmembers of complex backgrounds simultaneously, without estimating water IOPs or depth, as the network adaptively learns environmental parameters. Moreover, the linear mixture model (LMM) applied for unmixing shares the same physical significance with the bathymetric model, that is, the linear weighting of water and target spectra. The LMM enables the unmixing network to separate target signals from the background, thereby eliminating the influence of water on target spectra. The proposed nonlinear unmixing network for hyperspectral-UTD (NUN-UTD) approach considers underwater targets as mixed pixels and employs a signal separation-based method to obtain the abundance values of targets within pixels, thereby achieving nearshore UTD.

Existing HU models struggle to effectively adapt to the UTD task due to the following reasons.

1) *Lack of Prior Information Guidance and Constraints in the Unmixing Process:* Traditional unmixing algorithms are designed to address the mixed pixel in low-resolution HSI. However, when applied to specific target detection tasks, they require the guidance of prior target information. The lack of prior knowledge guidance impedes the direct transfer of conventional unmixing algorithms to UTD, as the endmembers learned by unconstrained or unguided networks often have no actual physical meaning [42].
2) *HU is Unable to Reliably Extract Weak Target Signals:* After undergoing absorption and scattering by the water, the underwater target signals are so weak that unmixing algorithms might fail to extract the endmembers associated with the targets.
3) *Nonlinear Interaction Between Underwater Substances Is Hard to Characterize:* The complex interaction mechanisms between underwater substances make a simple LMM insufficient for extracting underwater target signatures.

To address the aforementioned issues, we propose a nonlinear unmixing network for UTD (NUN-UTD), which aims to accurately separate the distorted target signal from the imagery while adapting to the complex nearshore underwater environment. Specifically, we propose an autoencoder-based nonlinear unmixing framework tailored to nearshore UTD. Inspired by Shen et al. [43], consider the underwater targets as weak signals submerged within the water spectrum. The HSI is fed into an encoder for feature representation, and the resulting abundance map is learned through the reconstruction task of the autoencoder. To preserve the weak signals of underwater targets, we employ a lower learning rate to fine-tune the network-initialized endmembers. This approach enables these endmembers to guide the learning of corresponding abundances during the unmixing process. In order to address the issue of nonlinear spectral interactions in underwater scenarios [28] and improve the model's adaptability to the environment, we employ a carefully designed additive post-nonlinear decoder [44]. Moreover, we propose a pseudomixed data training strategy to enable the unmixing network to better discriminate target and water spectra.

The contributions of this work are given as follows.

1) We constructed an open-source HSI dataset for nearshore UTD, named River Scene. The dataset and the NUN-UTD code will be available at https://github.com/Jiaxuan2021/NUN-UTD.
2) A prior-preserving strategy and a pseudomixed data training strategy are proposed for unmixing networks. This guides the autoencoder to learn a detection head embedded with prior target knowledge, thereby facilitating UTD.
3) We propose a nonlinear autoencoder unmixing network tailored for nearshore UTD. Extensive experiments are conducted on three real-world datasets (River Scene) and

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one simulated dataset, demonstrating that the proposed network achieves state-of-the-art detection performance.

The remainder of this article is organized as follows. Section II briefly introduces some preliminaries of this study. Then, we detail the NUN-UTD in Section III. In Section IV, the experimental results, ablation study, and parameter sensitivity of the model are presented. Finally, the conclusion of this article is drawn in Section V.

II. Autoencoder-Based HU

The autoencoder-based unmixing framework can simultaneously obtain endmembers and abundances in an unsupervised fashion. LMM forms the theoretical basis for unmixing methods based on autoencoder [44], [45] and has been widely used in most scenarios. Let $\mathcal{Y} \in \mathbb{R}^{r\times c\times B}$ represent the input HSI with $r \times c$ pixels and B spectral bands, and $\mathbf{Y} \in \mathbb{R}^{N\times B}$ denotes unfolding of $\mathcal{Y}$ with $N = r \times c$. According to the LMM, the input spectra matrix $\mathbf{Y}$ can be expressed as follows:

$$\mathbf{Y} = \mathbf{EA} + \mathbf{N}_{\text{noise}} \tag{1}$$

where $\mathbf{E} \in \mathbb{R}^{B\times m}$ denotes an endmember spectra matrix with m endmembers, $\mathbf{A} \in \mathbb{R}^{m\times N}$ represents the corresponding abundance coefficients, and additive noise is denoted by $\mathbf{N}_{\text{noise}} \in \mathbb{R}^{B\times N}$. Meanwhile, the abundance vector $\mathbf{a_j} \in \mathbb{R}^m$ from $\mathbf{A}$ for the jth pixel are required to satisfy two physical constraints, i.e., the abundance sum-to-one constraint (ASC) and the abundance nonnegativity constraint (ANC), as follows:

$$\sum_{i=1}^{m} \mathbf{a}_{ij} = 1$$
$$\mathbf{a}_{ij} \geq 0 \quad \forall i \in \{1, \ldots, m\}\ \forall j \in \{1, \ldots, N\}. \tag{2}$$

Deep autoencoder networks usually consist of an encoder and a decoder. The encoder encodes the input pixels into low-dimensional hidden features, which can be expressed as

$$\mathbf{A} = f_{\text{enc}}(\mathbf{Y}) = \delta\left(\mathbf{W}_{\text{enc}}^{\mathbf{T}}\mathbf{Y} + \mathbf{b}\right) \tag{3}$$

where f_{enc} represents the encoding function, with $\mathbf{W}_{\text{enc}}$ and $\mathbf{b}$ denoting the weights and bias of the encoder, and δ indicates an activation function. The output of encoder $\mathbf{A}$ is regarded as the computed abundance maps.

Meanwhile, the decoder uses the hidden features output by the encoder to reconstruct the input pixels; for LMM, this can be expressed as

$$\hat{\mathbf{Y}} = f_{\text{dec}}(\mathbf{A}) = \mathbf{W}_{\text{dec}}^{\mathbf{T}}\mathbf{A} \tag{4}$$

where f_{dec} represents the decoding function with $\mathbf{W}_{\text{dec}}$ denoting the weights of the decoder, i.e., the endmembers extracted by the autoencoder from the input. Note that in LMM, the decoder consists of only one linear layer without bias, indicating that the input pixels are reconstructed by directly multiplying the endmember matrix and abundance coefficients. Once the autoencoder is trained property, the network can obtain the abundance maps corresponding to the endmembers by minimizing the reconstruction loss, which is performed by

$$\text{Loss}_{\text{AE}} = \left\|\hat{\mathbf{Y}} - \mathbf{Y}\right\|^2. \tag{5}$$

III. Proposed Method

In this section, the proposed method is described in detail. Fig. 2 illustrates the proposed NUN-UTD framework, where the input consists of an HSI of a nearshore scene containing underwater targets, along with the corresponding target prior spectrum. Subsequently, the detector, which is the encoder, outputs the detection result of the underwater target. The objective of NUN-UTD is to learn a detector capable of discriminating between the background and target while simultaneously being an encoder that learns the target abundance. The training process of NUN-UTD consists of three parts: target prior spectrum preservation, pseudomixed data training, and pixel reconstruction of the autoencoder network. First, the weights of the linear part in the decoder are initialized using the extracted background endmembers and the incorporated target prior spectra. Then, the water spectra are obtained using the water mask and linearly mixed with the target prior spectra according to the LMM. This mixture is input into the nonlinear part of the decoder to obtain a nonlinear representation, yielding pseudomixed spectral training data with labels. Finally, the obtained pseudomixed spectra are used for classification, and the HSI is fed into a nonlinear unmixing network for unsupervised reconstruction training. During the testing stage, the input HSI is fed into the encoder to obtain detection results. The details of the proposed NUN-UTD are described in the following.

A. Target Prior Spectrum Preservation Scheme

In HSI, water spectra are significantly weaker compared to substances on land due to water's strong absorption and weak reflection of light. Meanwhile, nearshore UTD datasets often contain some land objects with high reflectance. Consequently, in nearshore scenarios, general unmixing algorithms or endmember extraction algorithms typically extract endmembers that only include high-reflectance background components, excluding the weak water spectral signals and the underwater target spectra. Therefore, the purpose of the target prior preservation scheme is to ensure that the endmembers extracted by unmixing network always contain the spectrum of the target. The abundance map corresponding to the target endmember is the detection result.

As shown in Fig. 3, the first four endmembers ($m = 4$) are extracted from the input hyperspectral cube using the vertex component analysis (VCA) algorithms [46]. Due to the strong interference of water spectra and the weak signal of the target spectrum with few corresponding pixels, it is difficult for VCA to directly extract the target spectrum. The extracted endmembers only contain the high-reflectance terrestrial objects. Therefore, we directly add the target prior spectrum to the last column of the endmember matrix extracted by VCA, which can be expressed as

$$\mathbf{E}_{\text{init}} = \text{Concat}\left[\mathbf{E}_{\text{VCA}}, \vec{\mathbf{e}}_{\text{prior}}\right] \in \mathbb{R}^{B\times(m+1)}, \quad m = 4$$
$$\mathbf{E}_{\text{init}} \rightarrow f_{\text{dec-linear}} \tag{6}$$

where $\mathbf{E}_{\text{VCA}} \in \mathbb{R}^{B\times m}$ represents endmembers extracted by VCA. $\vec{e}_{\text{prior}} \in \mathbb{R}^{B\times 1}$ is the target prior spectrum. $\mathbf{E}_{\text{init}}$ is used to initialize the weights of the linear part in the decoder, which

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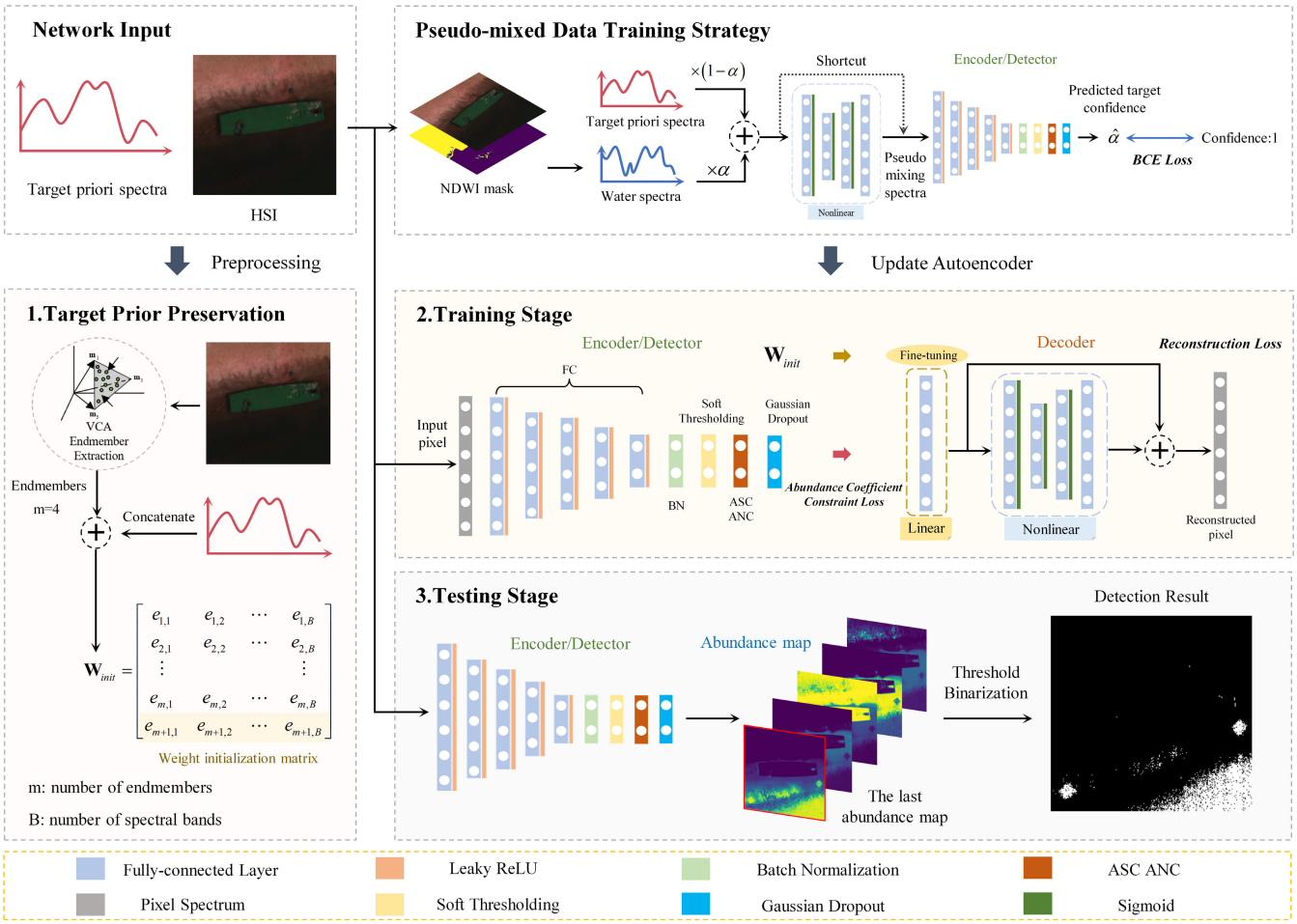


Fig. 2. Overview illustration of the proposed nonlinear unmixing autoencoder network for hyperspectral nearshore UTD. The LMM stands for linear mixture model.

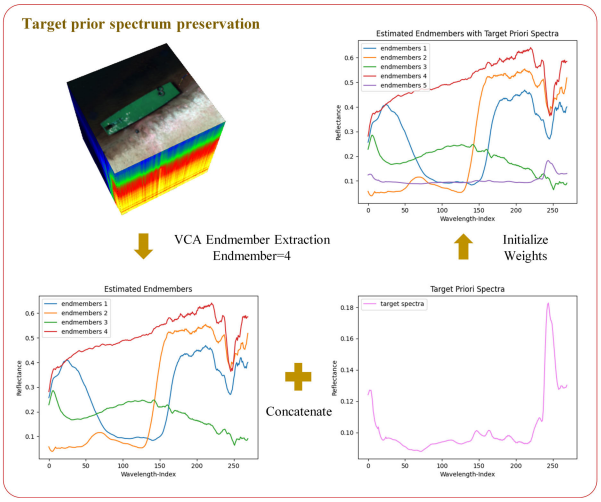


Fig. 3. Steps of target prior spectrum preservation. Background endmembers are extracted, and the target prior spectra are incorporated to generate the initial weights for the linear part of the decoder.

are then fine-tuned with a relatively small learning rate during training; this local learning rate is a ratio relative to the global learning rate

$$\gamma = \frac{\mathrm{lr}_{\mathrm{local}}}{\mathrm{lr}_{\mathrm{global}}} = \mathrm{ratio}. \tag{7}$$

When the environment is more complex and the spectral variability is more severe, the ratio should be set to a relatively larger value. In this way, the weak target signals would not be "overwhelmed" by the abundant background signatures during training, and the last abundance map corresponding to the target endmember is the detection result.

B. Hyperspectral Nonlinear Unmixing Autoencoder

In this work, the spectra of underwater targets are treated as nonlinear mixtures of target and water spectra. Therefore, we employ a nonlinear unmixing network to separate the target spectra and obtain their corresponding abundance maps, which represents the detection results. As illustrated in the upper half of Fig. 2, the autoencoder-based unmixing network comprises two components: an encoder and a decoder. The detailed summary of the encoder and the decoder in terms of layer type, input data shape, output data shape, and activation function is shown in Table I. It is worth noting that for unsupervised autoencoders, excessively deep network architectures can easily lead to overfitting, causing the model to focus too much on the reconstruction task, resulting in poor performance in abundance learning.

1) Abundance Learning Encoder: As shown by (3), the autoencoder network guides the encoder to learn a

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TABLE I
STRUCTURE OF THE NONLINEAR UNMIXING AUTOENCODER

Layer type	Input	Output	Activation function
Abundance learning encoder (m=4)			
FC	(512, 270)	(512, 18×m)	Leaky ReLU
FC	(512, 18×m)	(512, 9×m)	Leaky ReLU
FC	(512, 9×m)	(512, 6×m)	Leaky ReLU
FC	(512, 6×m)	(512, 3×m)	Leaky ReLU
FC	(512, 3×m)	(512, m+1)	Leaky ReLU
BN	(512, m+1)	(512, m+1)	—
Soft thresholding	(512, m+1)	(512, m+1)	—
ASC ANC	(512, m+1)	(512, m+1)	—
Gaussian droupout	(512, m+1)	(512, m+1)	—
Additive postnonlinear decoder			
FC (Linear)	(512, m+1)	(512, 270)	—
FC (Nonlinear)	(512, 270)	(512, 9×m)	Sigmoid
FC (Nonlinear)	(512, 9×m)	(512, 18×m)	Sigmoid
FC (Nonlinear)	(512, 18×m)	(512, 270)	Sigmoid
FC (Nonlinear)	(512, 270)	(512, 270)	—

low-dimensional representation, i.e., the abundance map, by minimizing the reconstruction loss. For target detection, the target abundance map is the part that we are most interested in, which is the output of the encoder. The encoder's first five layers with the Leaky ReLU activation function progressively reduce spectral dimensionality. Owing to the target prior spectrum preservation, the number of neurons in the last hidden layer exceeds endmembers by one, i.e., $(m+1)$. The output from that additional neuron corresponds to the target's confidence score, i.e., the detection map. Subsequently, batch normalization accelerates the process of learning, soft thresholding enhances the training stability, and the ASC ANC layer imposes nonnegativity and sum-to-one constraints on abundances, while Gaussian dropout bolsters robustness via random perturbations during propagation. Ultimately, the encoder outputs the abundance values, which are then fed as input into the decoder.

2) Additive Postnonlinear Decoder: The decoder uses the low-dimensional features from the encoder to reconstruct the input spectra. As shown in Fig. 2, the decoder consists of a linear part and a nonlinear part. The linear part represents the weighted sum of endmembers and their abundances, while the nonlinear part depicts the nonlinear interactions among spectra in HSI learned by the network. The final reconstructed result is the concatenation output of the two parts added to the output of the linear part for better representational capabilities, which is called the additive postnonlinear model and can be expressed as

$$\begin{aligned}\hat{\mathbf{Y}} &= f_{\text{dec-linear}}(\mathbf{A}) + f_{\text{dec-nonlinear}}(f_{\text{dec-linear}}(\mathbf{A})) \\ &= \mathbf{W}_{\text{linear}}^{\mathbf{T}}\mathbf{A} + \delta\left(\mathbf{W}_{\text{linear}}^{\mathbf{T}}\mathbf{A} + \mathbf{b}\right)\end{aligned} \tag{8}$$

where δ represents the sigmoid activation function. In addition, $\mathbf{A} \in \mathbb{R}^{N\times(m+1)}$ and $\mathbf{W}_{\text{linear}}^{\mathbf{T}} \in \mathbb{R}^{(m+1)\times B}$. As can be seen from (8), the linear part of the decoder does not use bias or activation function, following the LMM model. The nonlinear part, on the other hand, uses bias and sigmoid activation functions in order to achieve better nonlinear modeling capabilities. It should be noted that to safeguard the prior target spectrum, the linear portion of the decoder necessitates a fine-tuning approach, thus guaranteeing extraction of the underwater target spectrum.

The network outputs abundance maps after a sum-to-one constraint; using the standard $L1$ norm to achieve abundance sparsity constraint will be a constant (i.e., 1). Therefore, we adopt an approach to achieve abundance sparsity constraint by taking the logarithm of the reciprocal variance for abundance values. This approach promotes sparsity by penalizing deviations from a constant abundance, thereby enforcing each pixel to be represented by as few endmembers as possible. The loss function of the autoencoder can be mathematically described as follows:

$$\begin{aligned}\text{Loss}_{\text{AE}} &= \text{Loss}_{\text{Rec}} + \text{Loss}_{\text{Reg}} \\ &= \lambda\text{MSE}_{\text{Rec}} \\ &\quad + \mu\text{SUM}\left(\log\left(\frac{1}{\text{Var}(\mathbf{A}_{\text{col}}) + \text{eps}}\right)\right)\end{aligned} \tag{9}$$

$$\text{MSE}_{\text{Rec}}\left(\hat{\mathbf{Y}}, Y\right) = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2 \tag{10}$$

where Loss_{Rec} indicates the reconstruction loss and Loss_{Reg} refers to the regularization loss. MSE represents the mean square error loss as shown in (10), SUM stands for summation of vectors, λ and μ denote weighting coefficients, Var calculates the variance of the vector, and $\mathbf{A}_{\text{col}} \in \mathbb{R}^m$ indicates the abundance coefficients for all endmembers corresponding to each pixel spectrum. $\hat{\mathbf{Y}} \in \mathbb{R}^{N\times B}$ represents the reconstructed HSI, and $\mathbf{Y} \in \mathbb{R}^{N\times B}$ refers to the input HSI.

C. Pseudomixed Data Generation and Training

In HTD tasks, the targets of interest occupy only a small portion of the pixels in the HSI, leading to a severe class imbalance problem. Therefore, generating pseudotraining data for targets is a commonly used approach in HTD tasks. An LMM is often employed to synthesize target spectra [47], [48], as shown in the following equation:

$$L^{\text{target}} = \alpha \times L^{\text{prior}} + (1-\alpha)L^{\text{back}} \tag{11}$$

where α represents the abundance of the target, which is always randomly set to 0–1. L^{prior} is the target prior spectra, and L^{back} is the background spectra.

For HUTD, areas outside water regions can be easily distinguished using normalized difference water index (NDWI) masking, and these samples are considered simple samples that do not significantly contribute to network training. Therefore, we consider all water spectra as background, using random mixtures of the target prior spectra and water spectra as pseudomixed pixels. In this work, the weighting coefficient α is a random number within the range of 0.2–0.6, following a normal distribution, which simulates the proportion of the target spectrum within the actual underwater target pixels

$$L^{\text{pseudo}} = \alpha \times L^{\text{prior}} + (1-\alpha)L^{\text{water}}. \tag{12}$$

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The water spectrum L^{water} is obtained through the NDWI mask [49], as shown in the following equation:

$$\text{NDWI} = \frac{\text{I}_{\text{NIR}} - \text{I}_{\text{GREEN}}}{\text{I}_{\text{NIR}} + \text{I}_{\text{GREEN}}} \tag{13}$$

where I_{NIR} and I_{GREEN} represent the near-infrared and green bands in HSI, respectively. Higher values indicate water, while lower values represent land. Subsequently, an empirical threshold is set to obtain the water body mask through the following equation:

$$\mathbf{M}^{\text{water}} = \begin{cases} 255, & \text{NDWI} \leq \text{threshold} \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

Algorithm 1 Proposed NUN-UTD Method

Input: HSI: $\mathbf{Y}$, Target prior spectrum: L^{prior};
Result: Target abundance map;
E_{VCA} = VCA(HSI);
Initialize $f_{dec-linear}$ using E_{VCA} according to Eq.(6);
Generate NDWI mask and L^{water} according to Eq.(13) and Eq.(14);
Set the learning rate of model to lr_{global};
Set the learning rate of $f_{dec-linear}$ to lr_{local};
for $epoch = 1, 2, \ldots, n$ **do**
    Obtain L^{pseudo} via Eq.(12);
    Calculate pseudo loss through Eq.(15)-(18);
    Obtain $\hat{\mathbf{Y}}$ with Eq.(8);
    Calculate autoencoder loss through Eq.(9);
    Obtain total loss via Eq.(20);
    Back propagation;
end
Obtain detection map through Eq.(21);

However, for UTD, due to the nonlinear effects of water, the LMM cannot be simply used to synthesize underwater target spectra. In this way, we leverage the nonlinear network learned by the decoder to synthesize the pseudomixed data. In the proposed method, we input the LMM result into the nonlinear part of the decoder to obtain the nonlinear representation of the underwater target spectrum

$$L^{\text{nonlinear}} = f_{\text{dec}-\text{nonlinear}}\left(L^{\text{pseudo}}\right) \tag{15}$$

where $f_{\text{dec}-\text{nonlinear}}$ is the nonlinear network of the decoder. Then, the result is added to the LMM outcomes, which is consistent with the additive postnonlinear representation in (8)

$$L' = L^{\text{nonlinear}} + L^{\text{pseudo}}. \tag{16}$$

This approach yields the pseudomixed data for underwater targets. Feeding this data into the encoder part of the unmixing network enables obtaining the target abundance, which also represents the predicted target's confidence score

$$\hat{\alpha} = f_{\text{enc}}\left(L'\right) \tag{17}$$

where f_{enc} is the encoder network. Subsequently, the target abundance can be utilized to perform classification tasks, explained in the following manner:

$$\text{Loss}_{\text{pseudo}} = f_{\text{BCE}}\left(\hat{\alpha}, \alpha\right) \tag{18}$$

$$f_{\text{BCE}}\left(\hat{y}_i, y_i\right) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log\left(\hat{y}_i\right) + (1 - y_i)\log\left(1 - \hat{y}_i\right)\right] \tag{19}$$

where f_{BCE} indicates the binary cross-entropy loss, which is formulated by (19), where y_i and $\hat{y}_i$ represent the class label (0 or 1) and the predicted probability for the ith sample. The nonlinear network and encoder (detector) share parameters with their counterparts in the training stage, maintaining identical structures, as delineated in Table I. This approach enhances the network's discriminative capability between the target and background. Therefore, the overall loss function for the NUN-UTD network can be formulated as

$$\text{Loss} = \text{Loss}_{\text{pseudo}} + \text{Loss}_{\text{AE}}. \tag{20}$$

In summary, the proposed NUN-UTD method addresses the problem of nearshore HUTD by applying the target prior spectral preservation scheme and the pseudomixed data training strategy to the unmixing autoencoder. The NUN-UTD workflow is presented in Algorithm 1. During the test stage, the output target confidence map is given by the following expression:

$$\mathbf{A}_{\text{target}} = f_{\text{enc}}(\mathbf{Y})[-1] \tag{21}$$

where the value of -1 represents the final abundance map output by the encoder.

IV. Experimental Results and Analysis

To validate the effectiveness of the proposed NUN-UTD algorithm, we created an open-source nearshore UTD dataset called River Scene. In this section, we first introduce the datasets employed for the experiments, followed by a comparison of our proposed method against recent UTD algorithms and other traditional methods, along with an analysis of the experimental results. Subsequently, we verify the effectiveness of the major components of our proposed approach. Extensive experiments demonstrate that our proposed method achieves the best detection performance on three real-world datasets and a simulated dataset.

A. Datasets' Description

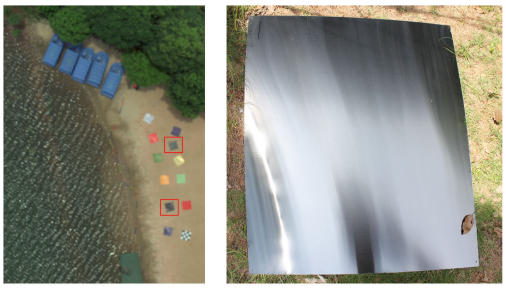
1) River Scene Datasets: As shown in Fig. 4, the River Scene dataset was captured by Headwall Nano-Hyperspec imaging sensor equipped on DJI Matrice 300 RTK unmanned aerial vehicle (UAV), and it was collected at the Qianlu Lake Reservoir in Liuyang (28°18′40.29″N, 113°21′16.23″E), Hunan Province, China, on July 31, 2021. The Headwall Nano-Hyperspec sensor features 270 spectral channels and 640 spatial channels, with a wavelength range of 400–1000 nm. The data were obtained along the shoreline of Qianlu Lake, including complex nearshore scenes. The underwater targets that we deployed were black metal plates with dimensions of approximately 1 × 1 m, as shown in Fig. 5(b). The targets were placed at depths ranging from 1 to 3 m in

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(a) (b) (c) (d)

Fig. 4. RGB image and ground truth of River Scene dataset and simulated dataset. (a) River Scene 1. (b) River Scene 2. (c) River Scene 3. (d) Simulated data.



(a) (b)

Fig. 5. (a) Scenarios for obtaining target prior spectra. (b) Placed underwater targets.

the river. Due to the similar color between the underwater targets and the water bottom, the underwater targets are not readily visible in the RGB imagery. Their specific locations are shown in the ground truth. The river scene images consist of the following pixel dimensions: River Scene 1 is 242 × 341 pixels, River Scene 2 is 255 × 261 pixels, and River Scene 3 is 137 × 178 pixels. The UAV flying height for River Scene 2 and River Scene 3 was 60 m, while the flying height for River Scene 1 was 100 m. River Scene 1 has a spatial resolution of 9.25 cm, while River Scene 2 and River Scene 3 have a spatial resolution of 5.55 cm. The reference target spectra for River Scene datasets were collected on land, as shown in Fig. 5(a). We utilized SpecView software to convert the HSI radiance values and perform Whiteboard correction. ENVI 5.3 is then employed to crop the HSI data and perform band selection. Radiance calibration, reflectance calibration, and Savitzky–Golay smoothing were applied to preprocess the datasets and their corresponding reference.

2) Simulated Dataset: To generate synthetic data, the first step is to create a background image using the water's IOPs. Next, a target spectrum is selected from the USGS spectral library, and the underwater target spectrum is then derived using bathymetric models [22]. Finally, the generated target spectrum is embedded within the background image [41]. We utilize the same synthetic turbid water dataset, as reported in [41], and introduce additive white Gaussian noise to simulate the interference from external environmental factors. In this dataset, the target prior spectra are represented by the sheet metal material spectrum, while the water's IOPs are measured and adopted from [50]. The dataset has a spatial resolution of 200 × 200 pixels, with wavelength coverage from 400 to 700 nm at 150 spectral bands. Furthermore, the target objects in the simulated data possess varying noise levels and depths.

B. Experimental Details

1) Evaluation Metrics: We employ quantitative evaluation using the area under the curve (AUC) metric and qualitative assessment through the receiver operating characteristic (ROC) curve to evaluate the effectiveness of the algorithms. These are widely adopted performance metrics in HTD. The ROC curve plots the true positive rate (TPR) on the y-axis against the false positive rate (FPR) on the x-axis. To evaluate the algorithm's performance, the TPR and FPR are computed for each classification threshold value between 0 and 1 applied to the output target mask. The formulas are given as follows:

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{22}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{TN} + \mathrm{FP}}. \tag{23}$$

The AUC metric is an effective indicator for evaluating the detection performance. It represents the area under the ROC curve, which can be computed by integrating the ROC curve over the coordinate axes. The AUC value ranges from 0 to 1, with a higher value indicating better detection capability. Formally, the AUC is defined as

$$\mathrm{AUC} = \int_{t=-\infty}^{\infty} y(t)dx(t) = \int_{t=-\infty}^{\infty} \mathrm{TPR}(t)d\mathrm{FPR}(t) \tag{24}$$

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TABLE II
RELEVANT PARAMETERS OF THE EXPERIMENTAL DATASETS

Dataset	γ	lr_{global}	λ	μ	Epochs	Batch Size	Pseudo Data	Binarization Threshold
Simulated Data	0.1	1e-4	10	1e-4	250	256	False	10%
River Scene 1	0.1	1e-4	10	1e-4	250	256	True	10%
River Scene 2	0.1	1e-4	10	1e-4	250	256	True	10%
River Scene 3	0.1	1e-4	10	1e-4	250	256	True	10%

TABLE III
AUC VALUES' COMPARISON FOR FOUR DATASETS

Dataset	ACE	CEM	OSP	SAM	GMM	VCA+FCLS	UTD-Net	TUTDF	HTD-IRN	NUN-UTD
Simulated Data	0.6980	0.4267	0.1461	0.9625	0.8997	0.9162	0.1101	0.9644	0.9762	**0.9808**
River Scene 1	0.5023	0.5455	0.7675	0.5469	0.3687	0.6090	0.3434	0.4879	0.9571	**0.9680**
River Scene 2	0.8743	0.8960	0.9041	0.3109	0.4150	0.1536	0.5220	0.2740	0.4939	**0.9697**
River Scene 3	0.4350	0.5806	0.8674	0.5470	0.3485	0.5239	0.7565	0.4754	0.8071	**0.9777**

where t represents the threshold applied to the target detection confidence map. A well-performing detector should exhibit a high AUC value, with the ROC curve tending toward the top-left corner of the plot.

2) Experimental Setups: In the proposed NUN-UTD framework, we employ the Adam optimizer with a weight decay of $1e^{-4}$ and a weight decay of 0 for the linear layers in the decoder. In addition, the negative slope of the Leaky ReLU activation function is set to 0.2, and the variance of the Gaussian dropout layer is 1.0. For the NDWI mask, the green and near-infrared bands used are 549.1280 and 941.3450 nm, respectively. The relevant experimental parameters are summarized in Table II.

In HTD tasks, class imbalance is a common issue, where targets occupy only a small number of pixels. To address this problem, according to [51], target detection networks achieve optimal performance when the number of background and target samples is balanced. For UTD in HSI, nonwater spectra are readily distinguishable and consequently less critical for network training. Thus, we consider the aquatic spectra as background. Therefore, we employ the pseudomixed data training strategy to synthesize underwater target spectra in quantities consistent with the water spectra. Meanwhile, water spectra can be obtained using the NDWI mask.

As the autoencoder training process is unsupervised and only utilizes the target prior spectra, we did not partition the data into distinct training and test sets. In other words, the training data and test data are the detection image itself. When performing the binarization threshold operation, we sort the obtained confidence scores in descending order and take the top portion of ranked pixels according to a specified ratio, which is presented in Table II. Since the simulated data were generated using the bathymetric model, the target and water spectra exhibit approximate linear mixing. Only the linear decoder is used for the simulated data in the experiments. The proposed network is implemented in PyTorch 1.13.1 on an Ubuntu 20.04 system, utilizing CUDA 11.7, Python 3.7, a GeForce RTX 2080 Ti GPU, and an Intel[1] Core[2] i7-8700 CPU @ 3.20 GHz, 32-GB RAM.

C. Comparison and Analysis of Experimental Results

Our proposed method is compared against four conventional signature detectors, including SAM [11], ACE [12], CEM [13], and OSP [52]. We also compared it with a Gaussian mixture model-based anomaly detector GMM. Meanwhile, we compared with two recently proposed deep learning-based UTD algorithms, namely, UTD-Net [41] and TUTDF [26]. It should be noted that the applicability of both methods is restricted to scenarios where the background consists solely of pure water. Therefore, in the comparative experiments, we applied water masking to the nearshore datasets before utilizing the TUTDF method for detection. An autoencoder-based HTD method, HTD-IRN [32], is also included in the comparison methods. Similar to the unmixing approach, HTD-IRN randomly initializes the endmember matrix, and the abundance matrix output from the encoder is multiplied by the endmember matrix to reconstruct the image. Ultimately, the target detection result is obtained from the encoder output. In order to demonstrate the performance improvement of the proposed method, we also compared it with the traditional spectral unmixing method VCA [46]. First, we extract the same number of endmembers using VCA and incorporate the target prior spectrum as a new endmember. Then, we use FCLS [53] to calculate the abundance map. Consequently, the abundance map corresponding to the target endmember represents the UTD results, i.e., VCA-FCLS.

Table III summarizes the detection performance of our proposed approach in comparison to the competing methods,

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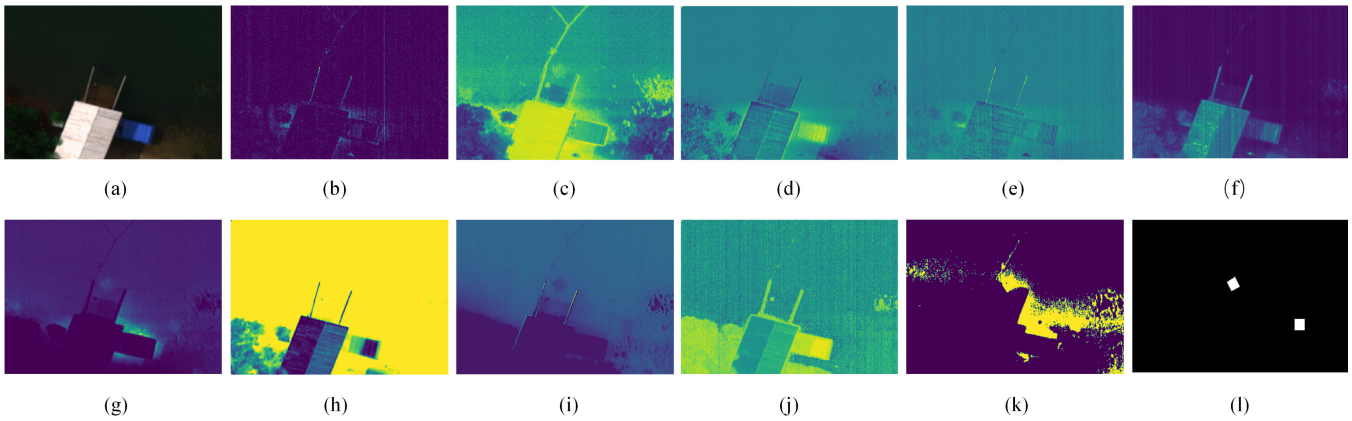


Fig. 6. Performance of the proposed and compared algorithms on River scene 1. (a) RGB image. (b) ACE. (c) SAM. (d) OSP. (e) CEM. (f) GMM. (g) HTD-IRN. (h) VCA+FCLS. (i) UTD-NET. (j) TUTDF. (k) Proposed method. (l) Ground truth.

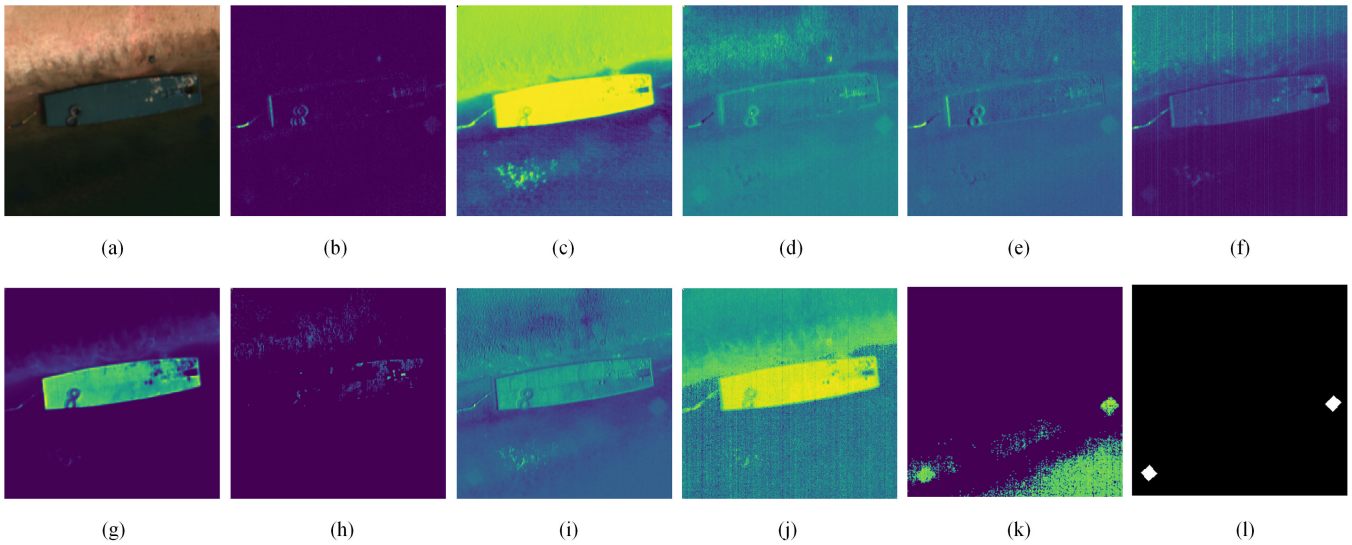


Fig. 7. Performance of the proposed and compared algorithms on River scene 2. (a) RGB image. (b) ACE. (c) SAM. (d) OSP. (e) CEM. (f) GMM. (g) HTD-IRN. (h) VCA + FCLS. (i) UTD-NET. (j) TUTDF. (k) Proposed method. (l) Ground truth.

evaluated on the River Scene datasets and the Simulated dataset. Figs. 6–9 present the detection maps of the proposed method and all the comparison methods. Fig. 10 shows the corresponding ROC curves. It is challenging to detect underwater targets for traditional HTD algorithms with severely distorted target prior spectra in complex nearshore backgrounds. Most algorithms exhibit significantly degraded performance in such scenarios. For the River Scene 1 dataset, apart from the NUN-UTD algorithm specifically designed for nearshore UTD, the OSP and HTD-IRN were able to yield weak detection scores at the target positions. HTD-IRN, which is similar to the unmixing method, achieved a high AUC score on the River Scene 1 dataset. As shown in Fig. 6, the detection result of UTD-Net is more focused on the water regions, incorrectly categorizing the underwater targets as part of the background. The other algorithms struggled to distinguish the spectral differences between the water and the underwater targets. The corresponding ROC curves in Fig. 10(a) demonstrate the superior detection performance of NUN-UTD.

River Scene 2 is relatively simple compared to River Scene 1 due to differences in lighting conditions and water turbidity. In this dataset, signature detectors, ACE, CEM, and OSP, achieve a good detection performance; they can produce weak confidence values at the locations of the targets. However, they struggled to effectively differentiate between nearshore land features and underwater targets. Deep learning-based methods, UTD-Net, TUTDF, and HTD-IRN, exhibited limited capability in effectively detecting underwater targets. Meanwhile, our proposed method achieves superior detection results. The ROC curves in Fig. 10(b) prove the above analysis.

Compared to River Scene 1 and River Scene 2, River Scene 3 is smaller in size and covers a more limited scene area. As shown in Fig. 8, the SAM algorithm incorrectly identified complex water bottom materials as targets. In contrast, the OSP and UTD-Net methods are able to output weak confidence values at the locations of the underwater targets. The NUN-UTD approach successfully detected the target locations but was also subject to interference from the complex water bottom, resulting in some false alarms. The ROC curves in

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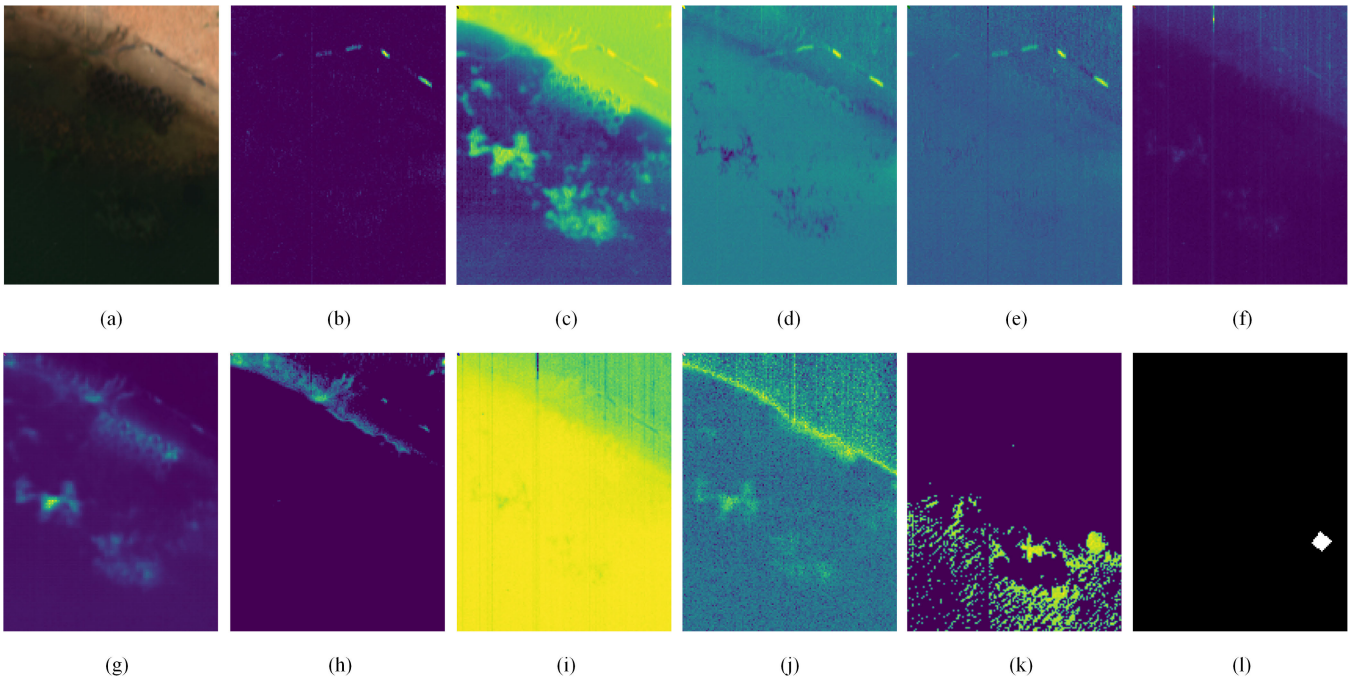


Fig. 8. Performance of the proposed and compared algorithms on River scene 3. (a) RGB image. (b) ACE. (c) SAM. (d) OSP. (e) CEM. (f) GMM. (g) HTD-IRN. (h) VCA + FCLS. (i) UTD-NET. (j) TUTDF. (k) Proposed method. (l) Ground truth.

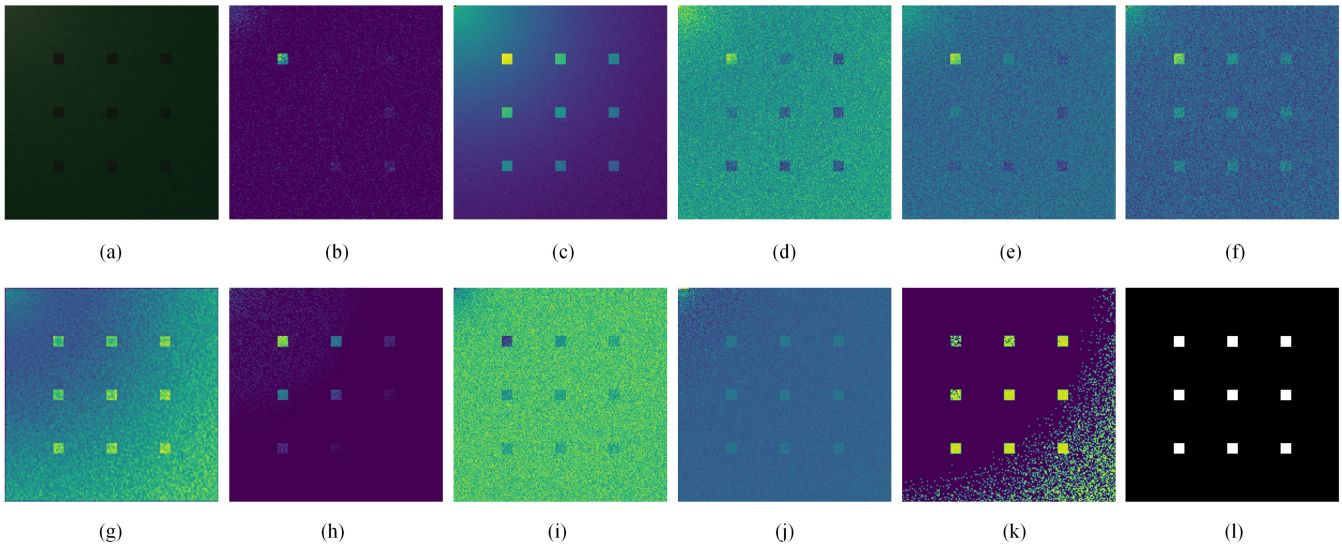


Fig. 9. Performance of the proposed and compared algorithms on a simulated dataset. (a) RGB image. (b) ACE. (c) SAM. (d) OSP. (e) CEM. (f) GMM. (g) HTD-IRN. (h) VCA + FCLS. (i) UTD-NET. (j) TUTDF. (k) Proposed method. (l) Ground truth.

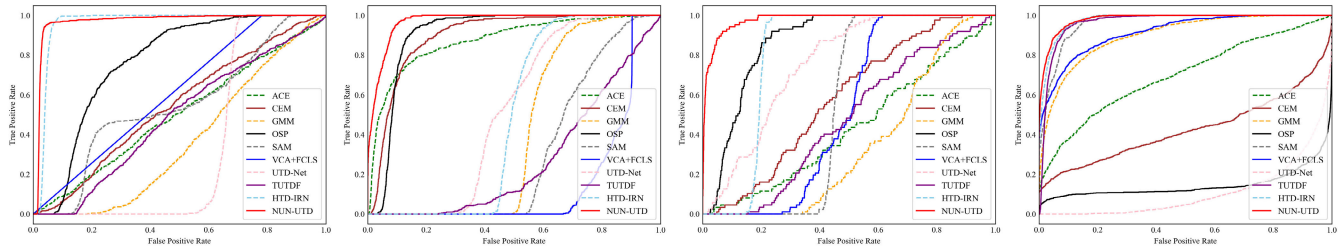
TABLE IV
AUC PERFORMANCE COMPARISON OF SEVERAL VARIANTS OF THE PROPOSED MODEL ON DIFFERENT DATASETS

Model	Prior Preservation	Nonlinear Decoder	Pseudo Data Training	River Scene 1	River Scene 2	River Scene 3
1	×	×	×	0.7716	0.9139	0.7022
2	✓	×	×	0.8264	0.9281	0.8191
3	×	✓	×	0.6265	0.9160	0.7551
4	✓	✓	×	0.7821	0.9194	0.8465
5	×	✓	✓	0.8637	0.9438	0.9759
6	✓	✓	✓	**0.9680**	**0.9697**	**0.9777**

Fig. 10(c) demonstrate the superior detection performance of our proposed method.

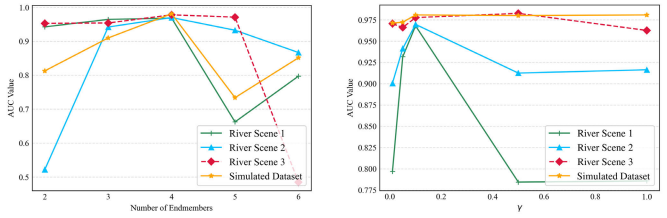
The simulated dataset is designed to emulate underwater target scenarios in deep-water environments. That is, the data

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(a) (b) (c) (d)

Fig. 10. ROC curves for River Scene datasets and simulated dataset. (a) River Scene 1. (b) River Scene 2. (c) River Scene 3. (d) Simulated data.



(a) (b)

Fig. 11. Sensitivity analysis of (a) number of endmembers extracted using VCA and (b) effect of the relative learning rate γ on four datasets.

only contain the pure water and the target. As shown in Fig. 9, SAM, GMM, VCA, HTD-IRN, and TUTDF achieve good performance in this dataset. Although the NUN-UTD approach is designed for nearshore UTD tasks, when the nonlinear pseudomixture data generation module is removed, the algorithm is still able to achieve superior detection performance on the simulated dataset. This demonstrates the potential of the NUN-UTD method for application in deep water scenarios. The ROC curves presented in Fig. 10(d) demonstrate that NUN-UTD outperforms the other comparison approaches.

D. Ablation Study

To evaluate the effectiveness of different components in NUN-UTD, we conducted a series of ablation studies on three real scene datasets. The effectiveness of each proposed module for improving detection performance is verified through a series of ablation experiments, and the specific experimental results are listed in Table IV. Given that targets occupy only a small fraction of pixels in the image, when adopting strategy variations, if certain target pixels are missed in the detection results, the AUC values may exhibit significant fluctuations.

1) Effectiveness of the Target Prior Spectrum Preservation: The prior preservation scheme in the NUN-UTD method is employed to incorporate target prior information into the reconstruction task. As illustrated in Table IV, compared with the baseline model, the model with prior preservation significantly boosts the detection performance on three datasets. This indicates that prior knowledge-guided unmixing models can be applied to enhance the performance of detection tasks. The experiments demonstrated that when prior preservation and nonlinear decoder were used in combination, the model was able to achieve better performance.

2) Effectiveness of the Nonlinear Decoder Unmixing: Our extensive experiments have demonstrated that the underwater target spectra, after undergoing absorption and reflection by the water, cannot be simply obtained through a linear weighted combination of the water spectrum and the target prior spectrum. This indicates that the distortion of the target spectrum in the nearshore environment is nonlinear. However, the nonlinear components of the network can lead to the autoencoder being easily overfitted. As illustrated in Table IV, the model with a nonlinear decoder has performance improvements in River Scene 2 and River Scene 3.

3) Effectiveness of Pseudomixed Data Training: The purpose of pseudomixed data training is to strengthen the network's ability to discriminate between water background and underwater targets. This strategy requires utilizing the nonlinear components of the decoder to obtain the nonlinear representation of the underwater target spectra and then using them to train the encoder. As presented in Table IV, the model utilizing pseudodata training and nonlinear decoder components can achieve performance results that are second only to the NUN-UTD approach.

E. Model Analysis

The interactions among spectra in underwater scenes are complex. To investigate whether an additive postnonlinear model is sufficient to express nonlinearity, we investigated the performance of deeper and more intricate networks on underwater object detection datasets. The experimental results are presented in Table V. The experimental results demonstrate that on the underwater dataset, more complex mixing models did not yield superior outcomes. The deeper networks' poor performance can be attributed to overfitting on the dataset's relatively simple scenes, leading to suboptimal results during testing.

F. Parameter Sensitivity

1) Number of Endmembers for VCA Initialization Weights: In this work, the linear weights of the decoder were initialized using the endmembers extracted by VCA, after incorporating the target prior spectral information. Using too few endmembers leads to poor reconstruction performance, while using too many endmembers can result in a decrease in the discriminative capability of the detector. To investigate the impact of the number of endmembers m on the

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TABLE V
AUC VALUES OF DIFFERENT NONLINEAR DECODERS ON HUTD DATASETS

Decoder	Deeper additive postnonlinear	Multi-level feature additive postnonlinear	Cascaded additive postnonlinear	Additive postnonlinear
River scene 1	0.8670	0.8882	0.9421	**0.9680**
River scene 2	0.9414	0.9171	0.9650	**0.9697**
River scene 3	0.9532	0.9655	0.9646	**0.9777**
Simulated data	0.9635	0.9492	0.9498	**0.9701**

φ_i denotes different nonlinear network in the decoder, x is the outcomes of the linear layer. The mathematical expressions for the four nonlinear decoders in the table are as follows: $\hat{y} = x + \varphi_2(\varphi_1(x))$, $\hat{y} = x + \varphi_1(x) + \varphi_2(\varphi_1(x))$, $\hat{y} = x + \varphi_2(x + \varphi_1(x))$, $\hat{y} = x + \varphi(x)$.

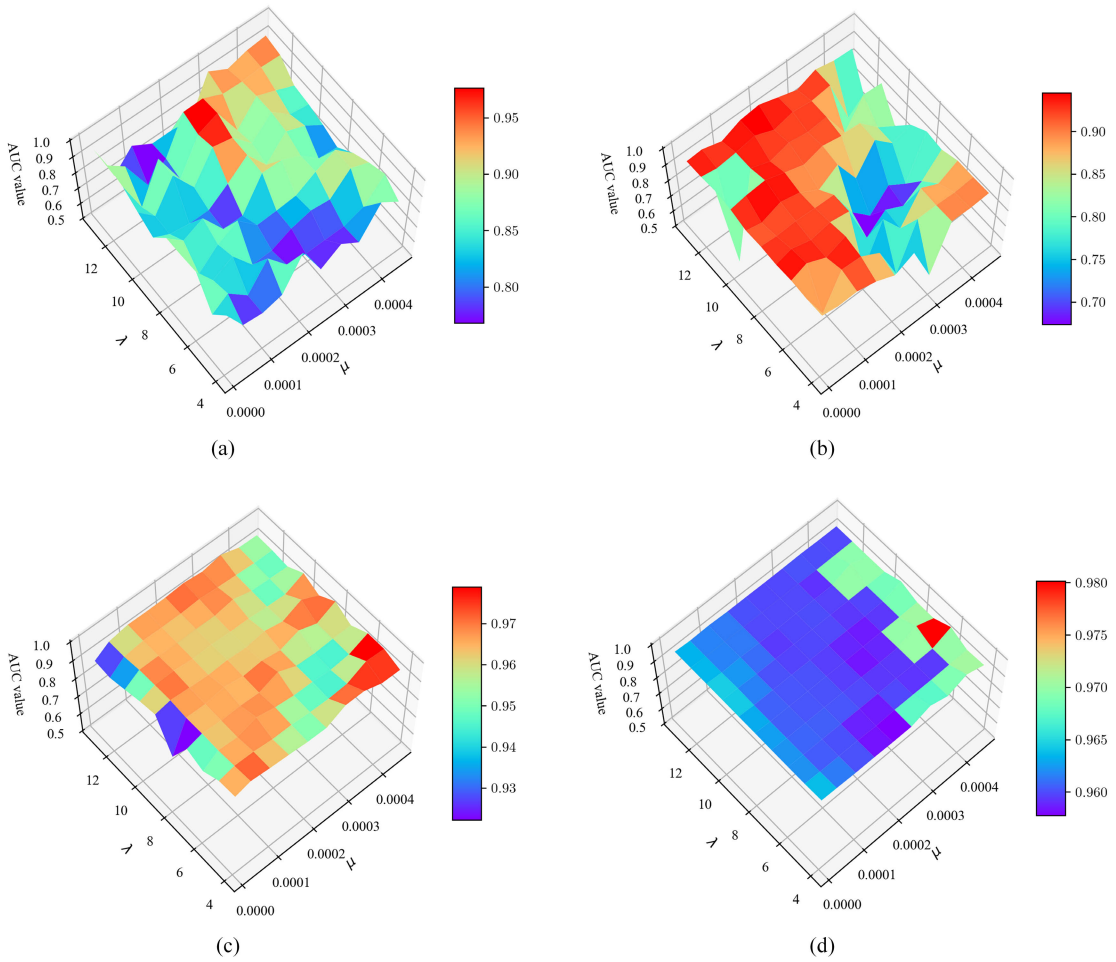


Fig. 12. AUC values for different weighting coefficients in the loss function. (a) River Scene 1. (b) River Scene 2. (c) River Scene 3. (d) Simulated data.

detection AUC value, we tested the algorithm's detection performance on the River Scene datasets and the simulated dataset with the endmembers set to 2–6, as shown in Fig. 11(a). The experiments demonstrated that when $m = 4$, NUN-UTD can achieve the best performance across the four datasets.

2) Relative Learning Ratio: In this work, the relative learning ratio γ is defined as the ratio of the learning rate for the linear fine-tuning component in the decoder to the overall learning rate. The value of γ is less than 1. This value is a hyperparameter, which is related to the complexity of the water environment. A larger γ indicates that the current endmembers are unable to effectively reconstruct the input HSI, implying a more complex environment. Conversely, a smaller γ suggests that the current endmembers can effectively represent the input HSI, indicating a simpler environment. When $\gamma = 1$, it means no constraint is imposed, allowing the network to freely learn the endmembers, which may result in the decoder learning unrealistic endmembers without physical significance. We set γ to 0.01, 0.05, 0.1, 0.5, and 1. The experimental results are shown in Fig. 11(b). In our detection datasets, the value of γ is set to 0.1.

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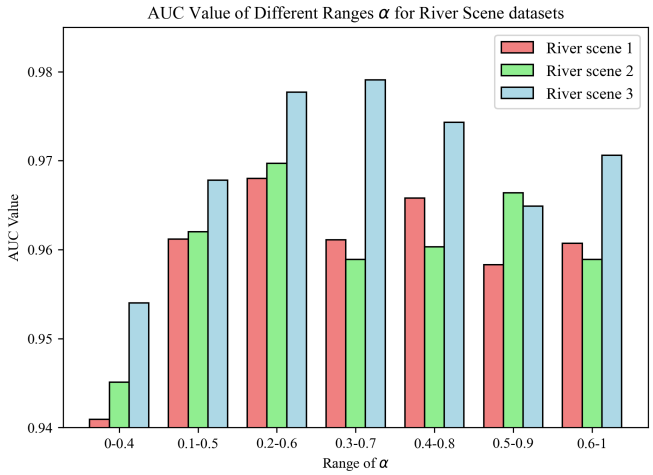


Fig. 13. Analysis of α in pseudodata learning on River Scene datasets.

3) Stability Analysis of the Weighting Coefficients in Loss Function: Equation (9) refers to the objective function of autoencoder. The parameters λ and μ are exploited to establish a tradeoff between reconstruction error and abundance sparsity constraint. Consequently, it is necessary to verify whether the detection performance is sensitive to these parameters. As illustrated in Fig. 12, we set the parameter μ to range from $1e^{-5}$ to $45e^{-5}$ and λ to range from 4 to 13, uniformly sampling ten combinations of these values. We then test 100 parameter configurations across the four datasets used in this study. To maintain consistent parameter settings that also yield favorable detection performance across the employed datasets, we ultimately set μ to $1e^{-4}$ and λ to 10.

4) Analysis of α in Pseudodata Learning: The pseudounderwater target spectra are synthesized using 12. In this equation, α is used to simulate the potential proportion of the target within underwater target pixels. To investigate the impact of the range of α on the experimental results, we selected values of α from 0 to 0.4, 0.1 to 0.5, 0.2 to 0.6, 0.3 to 0.7, 0.4 to 0.8, 0.5 to 0.9, and 0.6 to 1. The experimental results are shown in Fig. 13. It can be observed that NUN-UTD achieved the best performance when α falls within the range of 0.2–0.6.

V. Conclusion

For hyperspectral nearshore UTD, we propose a detection framework (NUN-UTD) based on a nonlinear unmixing autoencoder in this article. The key components of the method are an autoencoder-based unmixing network, target prior spectrum preservation, and pseudomixed data training. Specifically, we formulate the underwater target as a nonlinearly represented mixed pixel and leverage the target prior spectral information to guide the autoencoder's reconstruction task. Concurrently, to enhance the encoder's discriminative capability, we construct pseudounderwater target spectra using the nonlinear portion of the decoder and employ it as a classification task to guide the network training. Our proposed method is evaluated on River Scene datasets and simulated datasets that we constructed, demonstrating state-of-the-art detection performance.

In future work, we plan to investigate the challenge of false alarm reduction for nearshore UTD. Exploring techniques such as contrastive learning and unsupervised pretraining holds promising potential to enhance the network's ability to discriminate the target class from the confounding background signatures. Furthermore, we tend to explore methods for the simultaneous detection of multiple distinct target classes in the underwater environment. In addition, given the current scarcity of UTD datasets, we would like to construct datasets spanning diverse aquatic environments to enable a more comprehensive evaluation of algorithm performance.

References

[1] Z. Li, Z. Xie, P. Duan, X. Kang, and S. Li, "Dual spatial attention network for underwater object detection with sonar imagery," *IEEE Sensors J.*, vol. 24, no. 5, pp. 6998–7008, Mar. 2024.

[2] W. Wang and Y. Wang, "Underwater target detection system based on YOLO v4," in *Proc. 2nd Int. Conf. Artif. Intell. Inf. Syst.*, May 2021, pp. 1–5.

[3] G. Hu, K. Wang, Y. Peng, M. Qiu, J. Shi, and L. Liu, "Deep learning methods for underwater target feature extraction and recognition," *Comput. Intell. Neurosci.*, vol. 2018, pp. 1–10, Mar. 2018.

[4] P. J. Werdell et al., "An overview of approaches and challenges for retrieving marine inherent optical properties from ocean color remote sensing," *Prog. Oceanogr.*, vol. 160, pp. 186–212, Jan. 2018.

[5] L. Ge, G. Gao, and Z. Yang, "Study on underwater sea cucumber rapid locating based on morphological opening reconstruction and max-entropy threshold algorithm," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 32, no. 7, Jul. 2018, Art. no. 1850022.

[6] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.

[7] Z. Gong, X. Zhou, W. Yao, X. Zheng, and P. Zhong, "Hyper-DID: Hyperspectral intrinsic image decomposition with deep feature embedding," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5506714.

[8] F. Ferreira et al., "Heterogeneous marine robotic system for environmental monitoring missions," in *Proc. IEEE Underwater Technol. (UT)*, Mar. 2023, pp. 1–5.

[9] M. Ferrera, A. Arnaubec, K. Istenic, N. Gracias, and T. Bajjouk, "Hyperspectral 3D mapping of underwater environments," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 3696–3705.

[10] F. Foglini et al., "Underwater hyperspectral imaging for seafloor and benthic habitat mapping," in *Proc. IEEE Int. Workshop Metrology Sea, Learn. Measure Sea Health Parameters (MetroSea)*, Oct. 2018, pp. 201–205.

[11] F. A. Kruse et al., "The spectral image processing system (SIPS)-interactive visualization and analysis of imaging spectrometer data," *Remote Sens. Environ.*, vol. 44, pp. 145–163, May 1993.

[12] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.

[13] J. C. Harsanyi, *Detection and Classification of Subpixel Spectral Signatures in Hyperspectral Image Sequences*. Baltimore, MD, USA: Univ. Maryland, Baltimore County, 1993.

[14] L. Chen, J. Liu, S. Sun, W. Chen, B. Du, and R. Liu, "An iterative GLRT for hyperspectral target detection based on spectral similarity and spatial connectivity characteristics," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505811.

[15] C.-I. Chang, "Hyperspectral target detection: Hypothesis testing, signal-to-noise ratio, and spectral angle theories," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–23, 2021.

[16] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[17] R. Zhao, B. Du, and L. Zhang, "A robust nonlinear hyperspectral anomaly detection approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1227–1234, Apr. 2014.

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[18] Y. Zhang, D. Bo, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2015.

[19] J. Qi, W. Xue, Z. Gong, S. Zhang, A. Yao, and P. Zhong, "Hybrid sequence networks for unsupervised water properties estimation from hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3830–3845, 2021.

[20] Q. Dong, S. Shang, and Z. Lee, "An algorithm to retrieve absorption coefficient of chromophoric dissolved organic matter from ocean color," *Remote Sens. Environ.*, vol. 128, pp. 259–267, Jan. 2013.

[21] H. Zhan, Z. Lee, P. Shi, C. Chen, and K. L. Carder, "Retrieval of water optical properties for optically deep waters using genetic algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 5, pp. 1123–1128, May 2003.

[22] Z. Lee, K. L. Carder, C. D. Mobley, R. G. Steward, and J. S. Patch, "Hyperspectral remote sensing for shallow waters. I. A semianalytical model," *Appl. Opt.*, vol. 37, no. 27, pp. 6329–6338, Sep. 1998.

[23] S. Jay, M. Guillaume, and J. Blanc-Talon, "Underwater target detection with hyperspectral data: Solutions for both known and unknown water quality," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 4, pp. 1213–1221, Aug. 2012.

[24] S. Jay and M. Guillaume, "A novel maximum likelihood based method for mapping depth and water quality from hyperspectral remote-sensing data," *Remote Sens. Environ.*, vol. 147, pp. 121–132, May 2014.

[25] J. Qi et al., "A self-improving framework for joint depth estimation and underwater target detection from hyperspectral imagery," *Remote Sens.*, vol. 13, no. 9, p. 1721, Apr. 2021.

[26] Z. Li et al., "A transfer-based framework for underwater target detection from hyperspectral imagery," *Remote Sens.*, vol. 15, no. 4, p. 1023, Feb. 2023.

[27] Q. Li, J. Li, T. Li, Z. Li, and P. Zhang, "Spectral–spatial depth-based framework for hyperspectral underwater target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4204615.

[28] M. Zhao, M. Wang, J. Chen, and S. Rahardja, "Hyperspectral unmixing for additive nonlinear models with a 3-D-CNN autoencoder network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[29] Q. Jin, Y. Ma, F. Fan, J. Huang, X. Mei, and J. Ma, "Adversarial autoencoder network for hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 4555–4569, Aug. 2021.

[30] R. Liu, C. Lei, L. Xie, and X. Qin, "A novel endmember bundle extraction framework for capturing endmember variability by dynamic optimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505217.

[31] M. Wang, M. Zhao, J. Chen, and S. Rahardja, "Nonlinear unmixing of hyperspectral data via deep autoencoder networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1467–1471, Sep. 2019.

[32] D. Shen, X. Ma, W. Kong, J. Liu, J. Wang, and H. Wang, "Hyperspectral target detection based on interpretable representation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519416.

[33] M. Li, B. Yang, and B. Wang, "A coarse-to-fine scheme for unsupervised nonlinear hyperspectral unmixing based on an extended multilinear mixing model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521415.

[34] Y. Su, X. Xu, J. Li, H. Qi, P. Gamba, and A. Plaza, "Deep autoencoders with multitask learning for bilinear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8615–8629, Oct. 2021.

[35] M. Li, B. Yang, and B. Wang, "Spectral–spatial reweighted robust nonlinear unmixing for hyperspectral images based on an extended multilinear mixing model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5543917.

[36] Y. Su et al., "DAAN: A deep autoencoder-based augmented network for blind multilinear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512715.

[37] T. Fang, F. Zhu, and J. Chen, "Hyperspectral unmixing based on multilinear mixing model using convolutional autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507316.

[38] M. Li, B. Yang, and B. Wang, "EMLM-net: An extended multilinear mixing model-inspired dual-stream network for unsupervised nonlinear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509116.

[39] D. Zhu, B. Du, M. Hu, Y. Dong, and L. Zhang, "Collaborative-guided spectral abundance learning with bilinear mixing model for hyperspectral subpixel target detection," *Neural Netw.*, vol. 163, pp. 205–218, Jun. 2023.

[40] D. Zhu, B. Du, and L. Zhang, "Learning single spectral abundance for hyperspectral subpixel target detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, pp. 10134–10144, 2024.

[41] J. Qi, Z. Gong, W. Xue, X. Liu, A. Yao, and P. Zhong, "An unmixing-based network for underwater target detection from hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5470–5487, 2021.

[42] D. Hong et al., "Endmember-guided unmixing network (EGU-Net): A general deep learning framework for self-supervised hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6518–6531, Nov. 2022.

[43] X. Shen, H. Liu, J. Qin, F. Ge, and X. Zhou, "Toward weak signal analysis in hyperspectral data: An efficient unmixing perspective," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5534314.

[44] J. Chen, M. Zhao, X. Wang, C. Richard, and S. Rahardja, "Integration of physics-based and data-driven models for hyperspectral image unmixing: A summary of current methods," *IEEE Signal Process. Mag.*, vol. 40, no. 2, pp. 61–74, Mar. 2023.

[45] L. Qi, Z. Chen, F. Gao, J. Dong, X. Gao, and Q. Du, "Multiview spatial–spectral two-stream network for hyperspectral image unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502016.

[46] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[47] J. Jiao, Z. Gong, and P. Zhong, "Triplet spectralwise transformer network for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519817.

[48] D. Zhu, B. Du, and L. Zhang, "Two-stream convolutional networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6907–6921, Aug. 2021.

[49] S. K. McFeeters, "The use of the normalized difference water index (NDWI) in the delineation of open water features," *Int. J. Remote Sens.*, vol. 17, no. 7, pp. 1425–1432, May 1996.

[50] V. E. Brando, J. M. Anstee, M. Wettle, A. G. Dekker, S. R. Phinn, and C. Roelfsema, "A physics based retrieval and quality assessment of bathymetry from suboptimal hyperspectral data," *Remote Sens. Environ.*, vol. 113, no. 4, pp. 755–770, Apr. 2009.

[51] P. Hensman and D. Masko, "The impact of imbalanced training data for convolutional neural networks," Ph.D. dissertation, Dept. Comput. Sci., KTH Roy. Inst. Technol., Stockholm, Sweden, 2015.

[52] J. C. Harsanyi and C.-I. Chang, "Hyperspectral image classification and dimensionality reduction: An orthogonal subspace projection approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 4, pp. 779–785, Jul. 1994.

[53] D. C. Heinz, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.



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